

STAMBPL1 activates the GRHL3/HIF1A/VEGFA axis through interaction with FOXO1 to promote angiogenesis in triple-negative breast cancer

Huan Fang, Huichun Liang, Chuanyu Yang, Dewei Jiang, Qianmei Luo, Wenming Cao ,
Huifeng Zhang , Ceshi Chen 

Kunming Institute of Zoology, Chinese Academy of Sciences, Kunming, China • Kunming College of Life Sciences, University of Chinese Academy of Sciences, Kunming, China • The Department of Breast Medical Oncology, Zhejiang Cancer Hospital, Hangzhou, China • Department of Pharmacy, the First People's Hospital of Yunnan Province/the Affiliated Hospital of Kunming University of Science and Technology, Kunming, China • Academy of Biomedical Engineering, Kunming Medical University, Kunming, China • The Third Affiliated Hospital, Kunming Medical University, Kunming, China

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This article uncovers a new **important** role of STAMBPL1 in promoting angiogenesis in triple-negative breast cancer (TNBC) and elucidates the specific mechanisms by which it activates the GRHL3/HIF1 α /VEGFA axis through interaction with FOXO1. The finding that STAMBPL1 mediates GRHL3 transcription through the interaction with FOXO1 is novel. These experimental results corroborate each other, forming a **solid** foundation of evidence that supports the main findings of the article.

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Abstract

In the clinic, anti-tumor angiogenesis is commonly employed for treating recurrent, metastatic, drug-resistant triple-negative and advanced breast cancer. Our previous research revealed that the deubiquitinase STAMBPL1 enhances the stability of MKP-1, thereby promoting cisplatin resistance in breast cancer. In this study, we discovered that STAMBPL1 could upregulate the expression of the hypoxia-inducible factor HIF1 α in breast cancer cells. Therefore, we investigated whether STAMBPL1 promotes tumor angiogenesis. We demonstrated that STAMBPL1 increased *HIF1A* transcription in a non-enzymatic manner, thereby activating the HIF1 α /VEGFA signaling pathway to facilitate TNBC angiogenesis. Through RNA-seq analysis, we identified the transcription factor GRHL3 as a downstream target of STAMBPL1 that is responsible for mediating *HIF1A* transcription. Furthermore, we discovered that STAMBPL1 regulates *GRHL3* transcription by interacting with the transcription factor FOXO1. These findings shed light on the role and mechanism of

STAMBPL1 in the pathogenesis of breast cancer, offering novel targets and avenues for the treatment of triple-negative and advanced breast cancer.

Introduction

Triple-negative breast cancer (TNBC) is a subtype of breast cancer characterized by a lack of expression of estrogen receptor (ER), progesterone receptor (PR), and human epidermal growth factor receptor 2 (HER2). Despite representing only 10–15% of all breast cancer cases, it is associated with a greater risk of recurrence, metastasis, and resistance to chemotherapy, leading to a poorer prognosis than other types of breast cancer¹. Therefore, understanding the pathogenesis of this subtype of breast cancer and identifying effective treatment targets are key areas of research focus and challenge. In the clinic, in addition to surgery, radiotherapy, and chemotherapy, treatment approaches for TNBC also include the use of epidermal growth factor receptor (EGFR) inhibitors, poly (ADP-ribose) polymerase (PARP) inhibitors, immune checkpoint inhibitors, and anti-angiogenic therapies^{2,3}.

For solid tumors, the survival, proliferation, and invasion of cancer cells rely on surrounding blood vessels to provide nutrients and oxygen. Cancer cells can increase the synthesis and secretion of the vascular endothelial growth factor A (VEGFA), activate endothelial cells in neighboring blood vessels through paracrine pathways, and stimulate tumor angiogenesis⁴. Therefore, investigating the upstream molecular mechanisms that activate VEGFA in cancer cells offers new targets for inhibiting TNBC angiogenesis via disruption of this signaling pathway.

The deubiquitinase STAMBPL1 (also known as AMSH-LP) belongs to the AMSH family and cleaves K63-linked polyubiquitin chains⁵. STAMBPL1 stabilizes XIAP to inhibit apoptosis in prostate cancer cells⁶ and confers resistance to honokiol-induced apoptosis in various cancer cell types by stabilizing Survivin and c-FLIP⁷. Our previous research demonstrated that STAMBPL1 enhances cisplatin resistance in TNBC cells through the stabilization of MKP-1⁸. In this study, we discovered that, independent of its deubiquitinating enzyme activity, STAMBPL1 upregulates HIF1 α expression in TNBC cells. We aimed to investigate the molecular mechanism by which it upregulates HIF1 α and explore its role in tumor angiogenesis in TNBC.

FOXO1, a member of the forkhead box transcription factor family, regulates various physiological and pathological processes, such as cell proliferation, apoptosis, autophagy, and oxidative stress⁹. In breast cancer, FOXO1 enhances the stemness of cancer cells by promoting SOX2 transcription¹⁰ and confers resistance to chemotherapy drugs in basal-like breast cancer cells by activating *KLF5* transcription¹¹. Furthermore, FOXO1 plays a crucial role in angiogenesis by upregulating VEGFA transcription¹². However, the specific mechanism by which FOXO1 regulates VEGFA transcription remains unclear, and its role in tumor angiogenesis in TNBC requires further investigation.

In this study, we discovered that STAMBPL1 facilitates the transcriptional regulation of *GRHL3* by interacting with FOXO1, consequently enhancing *HIF1A* transcription through GRHL3 to activate the HIF1 α /VEGFA pathway. This results in increased endothelial cell activity via paracrine signaling, thereby promoting tumor angiogenesis in TNBC.

Methods

Cell lines and reagents

All cell lines used in this study, including HCC1806, HCC1937, and HEK293T cells, were purchased from ATCC (American Type Culture Collection, Manassas, VA, USA) and validated by short tandem repeat (STR) analysis, and these cell lines tested negative for mycoplasma contamination. HCC1806 and HCC1937 cells were cultured in RPMI 1640 medium supplemented with 5% FBS. HEK293T cells were cultured in DMEM (Thermo Fisher, Grand Island, USA) with 5% FBS at 37°C with 5% CO₂. Primary human umbilical vein endothelial cells (HUVECs) were maintained in an EGM-2 Bullet Kit (CC-3162, Lonza, USA). AS1842856 (Cat#HY-100596) and apatinib (Cat#HY-13342S) were purchased from MCE (New Jersey, USA).

Plasmid construction and stable STAMBPL1, GRHL3 and FOXO1 overexpression

We constructed the full-length *STAMBPL1*/*GRHL3*/*FOXO1* genes and then subcloned them into the pCDH lentiviral vector. The packaging plasmids (pMDLg/pRRE, pRSV-Rev, and pCMV-VSV-G) and the pCDH-*STAMBPL1*/*GRHL3*/*FOXO1* expression plasmid were cotransfected into HEK293T cells (2 × 10⁶ in 10 cm plates) to produce lentiviruses. Following transfection for 48 hours, the lentivirus was collected and used to infect HCC1806 and HCC1937 cells. Forty-eight hours later, puromycin (2 µg/ml) was used to screen the cell populations.

Stable knockdown of STAMBPL1 and GRHL3

The pSIH1-H1-puro shRNA vector was used to express *STAMBPL1*, *GRHL3* and luciferase (LUC) shRNAs. *STAMBPL1* shRNA#1, 5'-GGAGCATCAGAGATTGATA-3'; *STAMBPL1* shRNA#2, 5'-GCTGCTATGCCTGACCATA-3'; *GRHL3* shRNA#1, 5'-CCTTGAGCTTCCTCTATGA-3'; *GRHL3* shRNA#2, 5'-AGAGGAAGATGCGCGATGA-3'; *Luciferase* shRNA, 5'-CUUACGCUGAGUACUUCGA-3'; HCC1806 and HCC1937 cells were infected with lentivirus. Stable populations were selected via the use of 1 to 2 mg/mL puromycin. The knockdown effect was evaluated by Western blotting.

RNA interference

The siRNA target sequences used in this study were as follows: *STAMBPL1* siRNA#1, 5'-GGAGCATCAGAGATTGATA-3'; *STAMBPL1* siRNA#2, 5'-GCTGCTATGCCTGACCATA-3'; *GRHL3* siRNA#1, 5'-CCTTGAGCTTCCTCTATGA-3'; *GRHL3* siRNA#2, 5'-AGAGGAAGATGCGCGATGA-3'; *HIF1α* siRNA#1, 5'-AAGAGGTGGATATGTCTGG-3'; *HIF1α* siRNA#2, 5'-CGTCGAAAAGAAAAGTCTCTT-3'; *FOXO1* siRNA#1, 5'-CCCAGAUGCCUAUACAAAC-3'; and *FOXO1* siRNA#2, 5'-CTCAAATGCTAGTACTATTAG-3'. All siRNAs were synthesized by RiboBio (RiboBio, China) and transfected at a final concentration of 50 nM.

Western blotting (WB) and antibodies

The WB procedure was described in our previous study¹³. Anti-*STAMBPL1* (sc-376526) and anti-GAPDH (sc-25778) antibodies were purchased from Santa Cruz Biotechnology (Santa Cruz, CA, USA). Anti-*FOXO1* (#2880S) and anti-*HIF1α* (#36169S) antibodies were purchased from CST. Anti-β-actin (A5441) and anti-GST (G7781) antibodies were purchased from Sigma-Aldrich (St. Louis, MO, USA). The anti-Flag (ab205606) antibody was purchased from Abcam.

Real-time polymerase chain reaction (RT-qPCR)

Total RNA was extracted via TRIzol (Invitrogen, 15596026). One microgram of total RNA was reverse transcribed to cDNA according to the manufacturer's instructions for the HiScript II QRT SuperMix for qPCR Kit (Vazyme, R223-01). For quantitative PCR (RT-qPCR), the SYBR Green Select Master Mix system (Applied Biosystems, 4472908, USA) was used on an ABI-7900HT system (Applied Biosystems, 4351405). The primers used for PCR were as follows: 18S forward, 5'-CTCAACACGGGAAACCTCAC-3'; 18S reverse, 5'-CGCTCCACCAACTAAGAACG-3'; *HIF1α* forward, 5'-

AAGTCTGCAACATGGAAGGTAT-3'; HIF1 α reverse, 5'-TGAGGAATGGGTTACAAAATC-3'; VEGFA forward, 5'-TGAGGAATGGGTTACAAAATC-3'; VEGFA reverse, 5'-ATCTGCATGGTGATGTTGGA-3'; GLUT1 forward, 5'-TCGTGGCATCTCATCGCC-3'; GLUT1 reverse, 5'-CCGGTTCCTCGTTGCGGT-3'; GRHL3 forward, 5'-GGGGCTGAGGAATGCGATCT-3'; and GRHL3 reverse, 5'-AATTTGCCGTCCAGCTCCC-3'.

Cell proliferation and migration assays

To detect the proliferation of HUVECs, we used the Click-iT™ EdU Alexa Fluor™ 647 Imaging Kit (Invitrogen) according to the manufacturer's protocol. Briefly, HUVECs were seeded on coverslips (BD Biosciences) at 0.5×10^5 cells per well. The next day, the supernatants were discarded, and the cells were cultured with conditioned medium (CM). Six hours later, the cells were incubated with EdU in CM for 4 hours, followed by fixation and staining. For each sample, three random fields were observed via fluorescence microscopy, and the total numbers of cells and EdU-positive cells were counted. To detect the migration of HUVECs, we performed a wound-healing assay. Twenty-four hours after seeding, the supernatants of the HUVECs were discarded, and the cells were scratched and cultured with CM for 24 hours. Wound closure was imaged via microscopy. For each image, the gap width was analyzed via ImageJ.

Tube formation assays

HUVECs (1×10^4) in CM were seeded onto Matrigel (BD Biosciences)-coated μ -Slide angiogenesis plates (ibidiGmbH, Munich, Germany). At 6 hours after seeding, images were taken via microscopy and then analyzed with ImageJ. The total branch length was measured.

Chromatin immunoprecipitation assays

After the sample preparation was completed, the subsequent experimental steps were performed according to the instructions of the Simple ChIP (R) Plus Kit (CST, # 9005). The PCR primers for amplifying the region of interest on the HIF1 α gene promoter were as follows: 5'-GACTGACAGGCTTGAAGTTTATGC-3' and 5'-TGTTGCTGTAACTTCAAGGAAA-3', and the PCR primers for amplifying the region of interest on the GRHL3 gene promoter were as follows: 5'-TTCTATCCCTTCTGTGCTGACCA-3' and 5'-TGTGCCAGACCCTACTCTGGG-3'.

Dual-luciferase reporter assays

The DNA fragments HIF1 α and GRHL3 were amplified from MCF10A cell genomic DNA via PCR [template](#) and then cloned and inserted into the pGL3-Basic vector. HEK293T cells were seeded in 24-well plates and transfected with pCDH-GRHL3-3 $\times$ Flag or pCDH-FOXO1-3 $\times$ Flag and pGL3 luciferase reporter plasmids (both 600 ng/well) together with the pCMV-Renilla control (5 ng/well). After transfection for 48 h, the cell lysates were collected, and the luciferase activities were detected via the dual-luciferase reporter assay system (Promega, USA). For the WT HIF1 α promoter (with the GRHL3 binding motif), the primers used for PCR were as follows: forward, 5'-gctagccgggctcgagatctCACTGCGCTCCAGCCTG-3'; reverse, 5'-cagtaccggaatgccaagcttCCTCAGACGAGGCAGCACTG-3'. For the mutant HIF1 α promoter (without the GRHL3 binding motif), the primers used for PCR were as follows: forward, 5'-TCTTTCCTGAGGCCTTCTATATGCTTAT-3'; reverse, 5'-ATAAGCATATAGGAAGGCCTCAGGAAAAGA-3'. For the WT GRHL3 promoter (with the FOXO1 binding motif), the primers used for PCR were as follows: forward, 5'-gctagccgggctcgagatctATTAACAAGGGTGAAGAGGG-3'; reverse, 5'-cagtaccggaatgccaagcttTGGAGGTATACCTCAACAGGTGC-3'. For the mutant GRHL3 promoter (without the FOXO1 binding motif), the primers used for PCR were as follows: forward, 5'-CTCCCCACCAAACAAAGAAGGAGAACACCCC-3'; reverse, 5'-GGGGTGTCTCCTTCTTTGTTGGTGGGGGAG-3'.

Immunofluorescence staining

HEK293T cells plated on cell culture slides were transfected with the STAMBPL1 and FOXO1 expression plasmids. Two days after transfection, the cells were fixed in 3.7% polyformaldehyde at 4°C overnight. After being blocked with 5% BSA at room temperature for 1 hour, the cells were stained with anti-STAMBPL1 (mouse) and anti-FOXO1 (rabbit) antibodies at 4°C overnight. The cells were subsequently stained with both an Alexa Fluor647-labeled anti-mouse secondary antibody and a FITC-labeled anti-rabbit secondary antibody (ZSGB-Bio, Beijing, China) at room temperature for 1 hour. Nuclei were stained with DAPI (Biosharp, BL739A) for 15 min. Images were captured via a confocal microscope.

Immunoprecipitation and GST pull-down

For endogenous protein interaction, cell lysates were first incubated with anti-FOXO1 antibodies or rabbit IgG (2729S; Cell Signaling Technology) and then incubated with protein A/G magnetic beads (HY-K0202; MCE). For the GST pull-down assay, the cell lysates were directly incubated with Glutathione Sepharose 4B (10312185; Cytiva) overnight at 4°C. For the IP-Flag assay, cell lysates were directly incubated with anti-Flag magnetic beads (HY-K0207; MCE) overnight at 4°C. The precipitates were washed four times with 1 ml of lysis buffer, boiled for 10 minutes with 1×SDS sample buffer, and subjected to WB analysis.

Xenograft experiments

We purchased 5- to 6-week-old female BALB/c nude mice from SLACCAS (Changsha, China). Animal feeding and experiments were approved by the animal ethics committee of Kunming Institute of Zoology, Chinese Academy of Sciences. HCC1806-shLuc, HCC1806-shSTAMBPL1, or HCC1806-shGRHL3 cells and HCC1806-PCDH-Vector or HCC1806-PCDH-STAMBPL1 cells (1×10^6 in Matrigel (BD Biosciences, NY, USA)) were implanted into the mammary fat pads of the mice. When the tumor volume reached approximately 50 mm³, the nude mice were randomly assigned to the control or treatment groups (n = 4/group). The control group was given vehicle alone, and the treatment group received the FOXO1 inhibitor AS1842856 (10 mg/kg), the VEGFR inhibitor apatinib (50 mg/kg), and the FOXO1 inhibitor AS1842856 combined with the VEGFR inhibitor apatinib via intragastric administration every two days for 20 days. The tumor volume was calculated as follows: tumor volume was calculated by the formula $(\pi \times \text{length} \times \text{width}^2)/6$. The maximal tumor size permitted by the animal ethics committee of Kunming Institute of Zoology, Chinese Academy of Sciences, and the maximal tumor size in this study was not exceeded.

Immunohistochemical staining

The xenograft tumor tissues were fixed in 3.7% formalin solution. The immunohistochemistry was performed on 4-μm-thick paraffin sections after pressure-cooking for antigen retrieval. An anti-CD31 primary antibody (1:400, Abcam, ab28364) was used. After 12 h, the slides were washed three times with PBS and incubated with secondary antibodies (hypersensitive enzyme-labeled goat anti-mouse/rabbit IgG polymer (OriGene, China) at room temperature for 20 min, DAB concentrated chromogenic solution (1:200 dilution of concentrated DAB chromogenic solution)), counterstained with 0.5% hematoxylin, dehydrated with graded concentrations of ethanol for 3 min each (70%–80%–90%–100%), and finally stained with dimethyl benzene. The immunostained slides were evaluated via light microscopy, and the number of microvessels with positive CD31 expression was counted.

Statistical analysis

All the graphs were created via GraphPad Prism software version 8.0. Comparisons between two independent groups were assessed via two-tailed Student's *t* tests. One-way analysis of variance with least significant differences was used for multiple group comparisons. *P* values of < 0.05, 0.01

or 0.001 were considered to indicate statistically significant results, and comparisons that were significant at the 0.05 level are indicated by *, those at the 0.01 level are indicated by **, and those at the 0.001 level are indicated by ***.

Results

STAMBPL1 upregulates HIF1 α in a non-enzymatic manner in TNBC cells

Our previous research revealed that STAMBPL1, a deubiquitinase, promotes cisplatin resistance in TNBC by stabilizing MKP1⁸. To further investigate the role and mechanism of STAMBPL1 in TNBC, we discovered that the knockdown of STAMBPL1 can inhibit hypoxia-induced HIF1 α expression (Fig. 1A) and suppress the transcription of the HIF1 α downstream genes *VEGFA* and *GLUT1* (Fig. 1B-E). Under normoxic conditions, both STAMBPL1 and its enzymatically mutated variant (E292A) can upregulate the protein expression of HIF1 α (Fig. 1F). These findings suggest that STAMBPL1 enhances the expression of HIF1 α in breast cancer cells through a non-DUB enzyme activity mechanism.

STAMBPL1 promotes *HIF1A* transcription and activates the HIF1 α /VEGFA axis

To elucidate how STAMBPL1 upregulates HIF1 α , we examined the transcription levels of *HIF1A*. These results indicate that knocking down STAMBPL1 significantly inhibits the transcription of *HIF1A* (Fig. 2A-C). STAMBPL1 overexpression increased the transcription of *HIF1A* (Fig. 4E and SFig. 3E) and *VEGFA* (Fig. 2E and 2G). The ability of STAMBPL1 to induce *VEGFA* transcription was blocked by HIF1 α knockdown (Fig. 2D-G). These findings suggested that STAMBPL1 activated the HIF1 α /VEGFA axis through enhancing the transcription of *HIF1A*.

STAMBPL1 in TNBC cells enhances the activity of HUVECs and promotes TNBC angiogenesis

The conditioned medium (CM) from TNBC cells with STAMBPL1 knockdown inhibited the proliferation (Fig. 3A-B, and SFig. 1A-B), migration (Fig. 3C-D, and SFig. 1C-D), and tube formation (Fig. 3E-F, and SFig. 1E-F) of HUVECs. When STAMBPL1 was overexpressed in TNBC cells, the conditioned medium of HCC1806 and HCC1937 cells promoted the ability of HUVECs to proliferate (SFig. 2A and 2D), migrate (SFig. 2B and 2E), and form tubes (SFig. 2C and 2F), which could be reversed by knocking down HIF1 α in TNBC cells. These findings suggest that STAMBPL1 activates the HIF1 α /VEGFA axis in TNBC cells, leading to enhanced abilities of HUVECs through a paracrine pathway. Knocking down STAMBPL1 inhibited the growth of HCC1806 xenografts in mice (Fig. 3G-I) and decreased the number of microvessels in tumor tissue (Fig. 3J-K), indicating that STAMBPL1 promoted tumor angiogenesis.

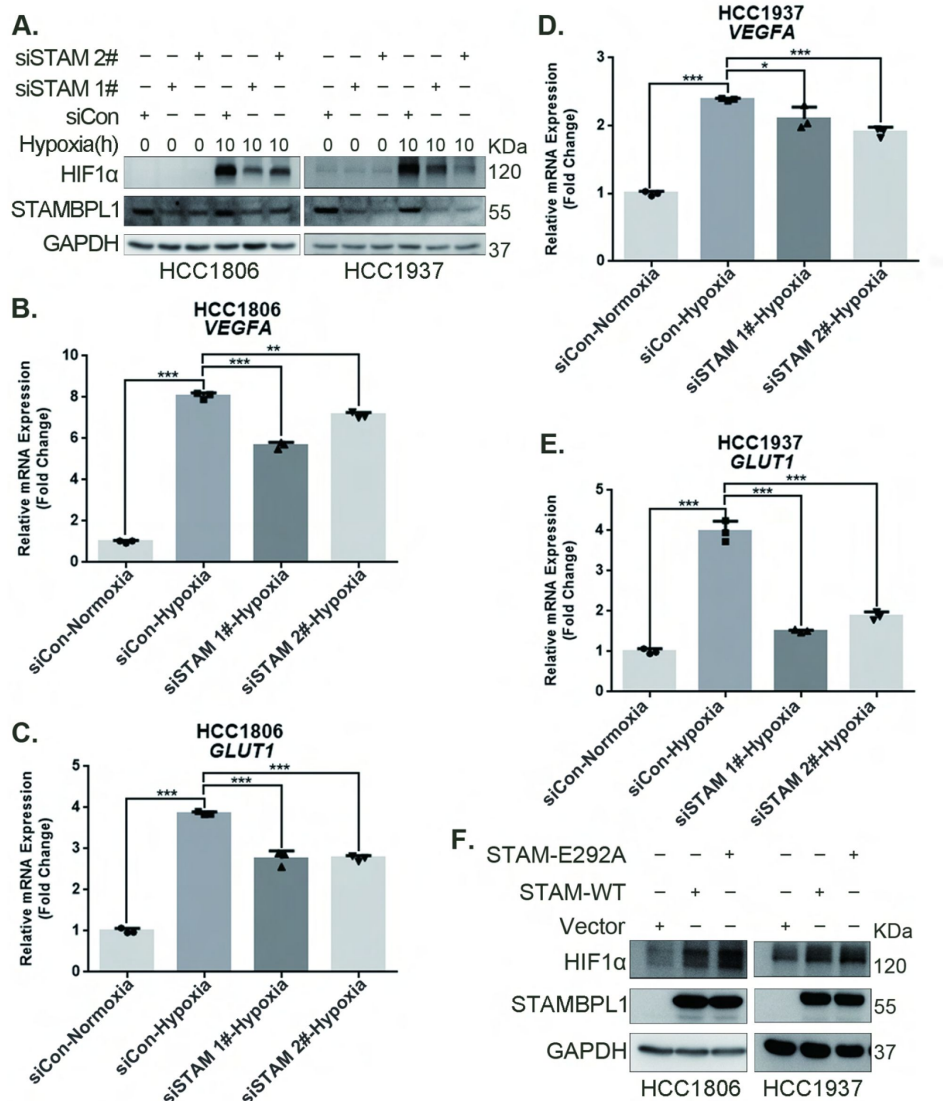


Figure 1.

STAMBPL1 upregulates HIF1 α in a non-enzymatic manner in TNBC cells.

(A) After STAMBPL1 was knocked down in HCC1806 and HCC1937 cells and then subjected to hypoxia for 10 h, STAMBPL1 knockdown downregulated the protein level of HIF1 α . (B-C) RNA samples were collected after 10 h of hypoxia treatment following the knockdown of STAMBPL1 in HCC1806 cells. RT-qPCR experiments revealed that the knockdown of STAMBPL1 downregulated the mRNA levels of the HIF1 α downstream targets *VEGFA* and *GLUT1*. (D-E) RNA samples were collected after 10 hours of hypoxia treatment following the knockdown of STAMBPL1 in HCC1937 cells. RT-qPCR experiments revealed that the knockdown of STAMBPL1 downregulated the mRNA levels of the HIF1 α downstream targets *VEGFA* and *GLUT1*. (F) Overexpression of STAMBPL1/STAMBPL1-E292A upregulated HIF1 α protein expression in HCC1806 and HCC1937 cells stably overexpressing STAMBPL1/STAMBPL1-E292A under normoxia. *: $P < 0.05$, **: $P < 0.01$, ***: $P < 0.001$, t test.

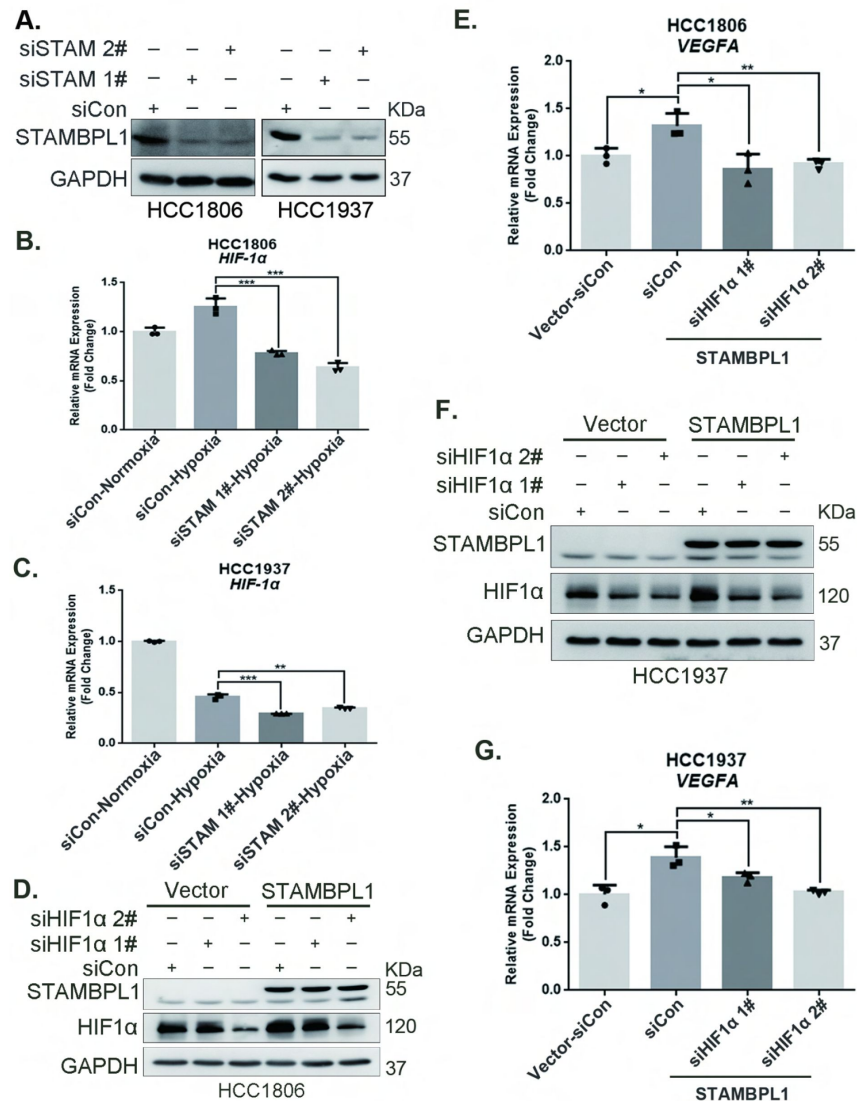


Figure 2.

STAMBPL1 promotes *HIF1A* transcription and activates the HIF1α/VEGFA axis.

(A) Western blot analysis was performed to confirm the knockdown of STAMBPL1 in HCC1806 and HCC1937 cells. (B) RNA samples were collected after 10 h of hypoxia treatment following the knockdown of STAMBPL1 in HCC1806 cells. RT-qPCR experiments revealed that the knockdown of STAMBPL1 could downregulate the mRNA levels of HIF1α. (C) RNA samples were collected after 10 hours of hypoxia treatment following the knockdown of STAMBPL1 in HCC1937 cells. RT-qPCR experiments revealed that the knockdown of STAMBPL1 downregulated the mRNA levels of HIF1α. (D) HCC1806 cells stably overexpressing STAMBPL1 were used to knock down HIF1α via siRNA. The protein was then collected and subjected to Western blotting experiments. The results showed that the overexpression of STAMBPL1 promoted the expression of HIF1α, which could be rescued by the knockdown of HIF1α. (E) HIF1α was knocked down in HCC1806 cells stably overexpressing STAMBPL1 via siRNA. The RNA was then collected and subjected to RT-qPCR experiments. The results showed that the overexpression of STAMBPL1 promoted the expression of the VEGFA mRNA, which could be rescued by the knockdown of HIF1α. (F) In HCC1937 cells stably overexpressing STAMBPL1, HIF1α was knocked down via siRNA, and protein was collected. Western blotting experiments revealed that the overexpression of STAMBPL1 promoted HIF1α expression, which could be rescued by the knockdown of HIF1α. (G) In HCC1937 cells stably overexpressing STAMBPL1, HIF1α was knocked down via siRNA, and RNA was collected. RT-qPCR experiments revealed that the overexpression of STAMBPL1 promoted VEGFA mRNA expression, which could be rescued by the knockdown of HIF1α. *: $P < 0.05$, **: $P < 0.01$, ***: $P < 0.001$, t test.

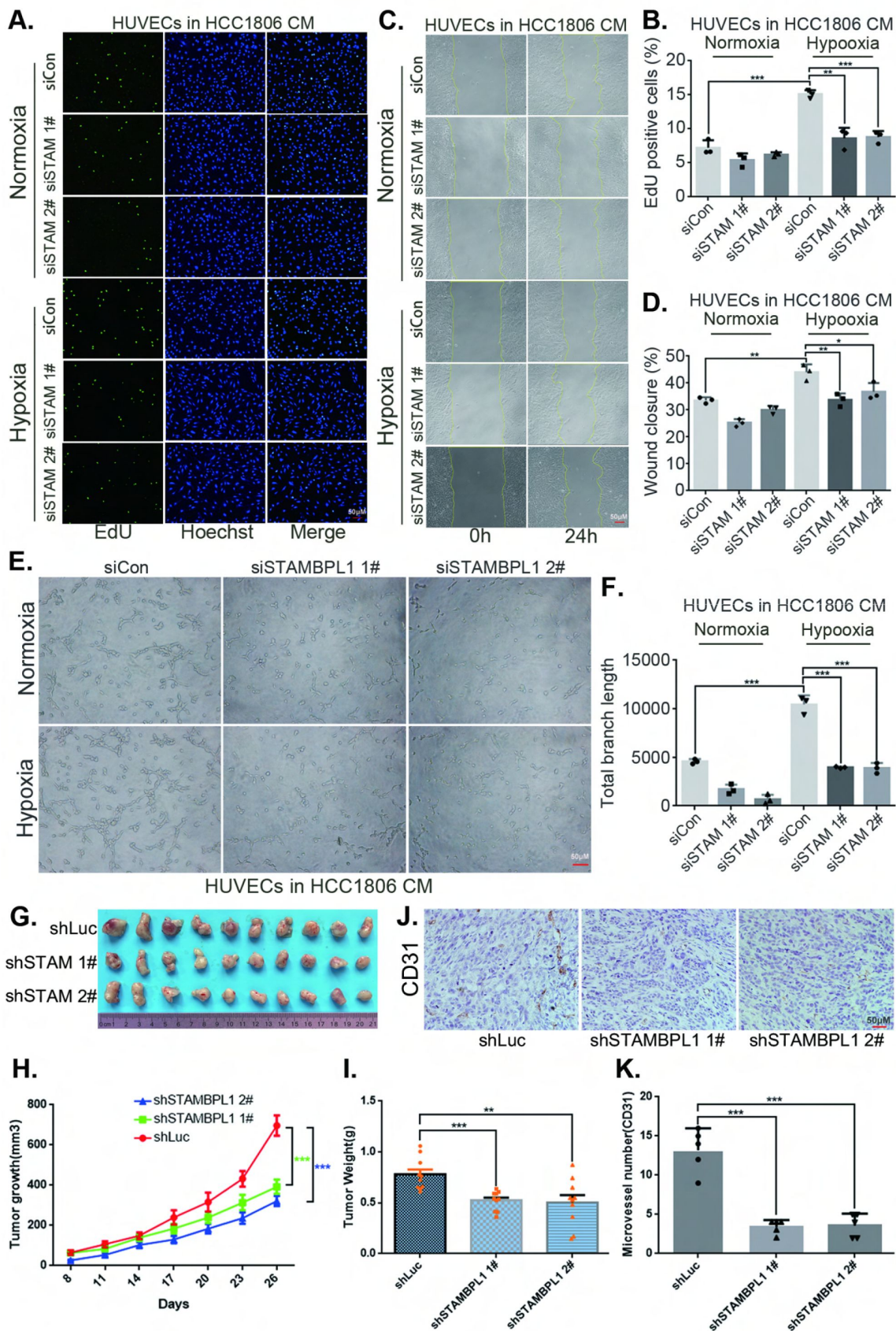


Figure 3.

STAMBPL1 in TNBC cells enhances the activity of HUVECs and promotes TNBC angiogenesis.

(A) In HCC1806 cells, STAMBPL1 was knocked down via siRNA, and the cells were then subjected to hypoxia for 24 hours. The conditioned medium (CM) collected from these cells was used to treat HUVECs. EdU assays revealed that the conditioned medium after STAMBPL1 knockdown inhibited the proliferation of HUVECs. (B) Statistical analysis of the EdU assay results. (C) In HCC1806 cells, STAMBPL1 was knocked down via siRNA, and the cells were then subjected to hypoxia for 24 hours. The CM collected from these cells was used to treat HUVECs. Wound healing assays demonstrated that the conditioned medium after STAMBPL1 knockdown inhibited the migration of HUVECs. (D) Statistical analysis of the wound healing assay results. (E) In HCC1806 cells, STAMBPL1 was knocked down via siRNA, and the cells were then subjected to hypoxia for 24 hours. The CM collected from these cells was used to treat HUVECs. A tube formation assay revealed that the conditioned medium after STAMBPL1 knockdown inhibited the tube formation of HUVECs. (F) Statistical analysis of the tube formation assay results. (G) The growth of TNBC xenograft tumors was evaluated by photographing the tumors in nude mice to assess the role of STAMBPL1. Knockdown of STAMBPL1 significantly inhibited tumor growth. (H-I) The growth and weight of the transplanted tumors in the nude mice were statistically analyzed, and the STAMBPL1sh#1 and STAMBPL1sh#2 groups were compared with the shLuc group. (J) An immunohistochemical assay was used to detect the expression of the angiogenesis marker CD31 in xenograft tumors. (K) The number of microvessels in the immunohistochemical experiments was statistically analyzed. *: $P < 0.05$, **: $P < 0.01$, ***: $P < 0.001$, t test.

STAMBPL1 promotes *HIF1A* transcription via the upregulation of GRHL3

By silencing the *STAMBPL1* gene in HCC1806 cells subjected to 10 hours of hypoxia, we performed RNA-seq analysis to investigate the mechanism by which STAMBPL1 promotes *HIF1A* transcription. We found that silencing of STAMBPL1 resulted in the downregulation of 27 genes (of which only 18 were annotated), with *GRHL3* being the sole gene encoding a transcription factor (Fig. 4A and SFig. 3A). Silencing of STAMBPL1 inhibited *GRHL3* transcription in both HCC1806 and HCC1937 cells (Fig. 4B and SFig. 3B). Conversely, the overexpression of STAMBPL1 promoted *GRHL3* transcription (Fig. 4D and SFig. 3D). Furthermore, the stimulatory effects of STAMBPL1 on the HIF1 α /VEGFA axis (Fig. 4C-F and SFig. 3C-F) and on HUVECs (Fig. 4G-I and SFig. 3G-I) were reversed by GRHL3 knockdown. These findings suggest that STAMBPL1 activates the HIF1 α /VEGFA axis by upregulating GRHL3.

GRHL3 enhances *HIF1A* transcription by binding to its promoter

On the basis of the GRHL binding motif (Fig. 5A), a GRHL binding sequence was identified in the *HIF1A* promoter (Fig. 5B). ChIP and luciferase assays revealed that GRHL3 bound to the *HIF1A* promoter (Fig. 5D) and increased its activity (Fig. 5E). Knockdown of GRHL3 (Fig. 5G and SFig. 4B) resulted in decreased expression of HIF1 α at both the mRNA (Fig. 5H and SFig. 4C) and protein levels (Fig. 5F and SFig. 4A), leading to suppressed *VEGFA* transcription (Fig. 5I and SFig. 4D), indicating that GRHL3 promotes *HIF1A* transcription by binding to its promoter.

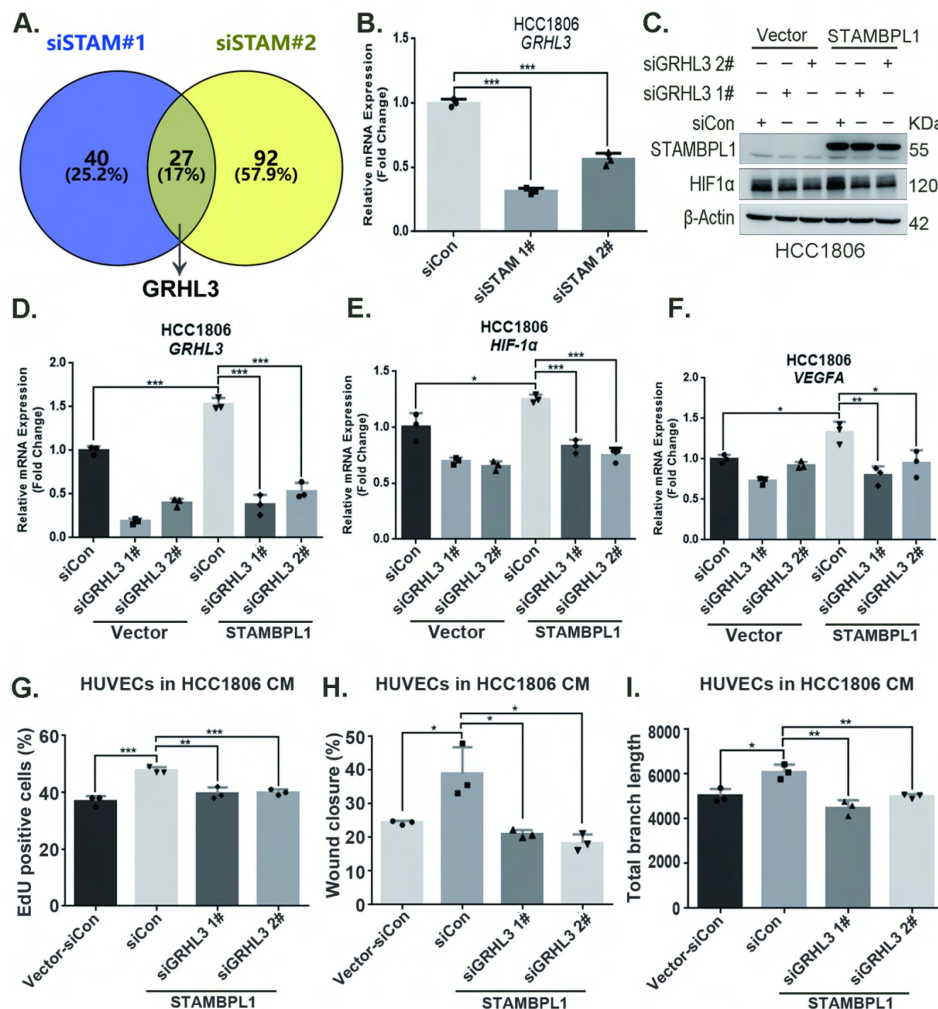


Figure 4.

STAMBPL1 promotes *HIF1A* transcription via the upregulation of GRHL3.

(A) Venn diagram showing the overlap of differentially expressed genes identified via RNA-seq analysis. (B) RT-qPCR experiments demonstrated that the knockdown of STAMBPL1 resulted in decreased mRNA levels of GRHL3 in HCC1806 cells. (C) In HCC1806 cells stably overexpressing STAMBPL1, GRHL3 was knocked down via siRNA, after which the protein was collected. Western blotting experiments revealed that the overexpression of STAMBPL1 promoted the expression of HIF1α, which could be rescued by the knockdown of GRHL3. (D-F) In HCC1806 cells stably overexpressing STAMBPL1, GRHL3 was knocked down via siRNA, and RNA was subsequently collected. RT-qPCR experiments revealed that the overexpression of STAMBPL1 upregulated the mRNA expression of GRHL3, HIF1α and its downstream target VEGFA, which could be rescued by the knockdown of GRHL3. (G) Statistical analysis of the EdU assay results. (H) Statistical analysis of the wound healing assay results. (I) Statistical analysis of the tube formation assay results. *: $P < 0.05$, **: $P < 0.01$, ***: $P < 0.001$, t test.

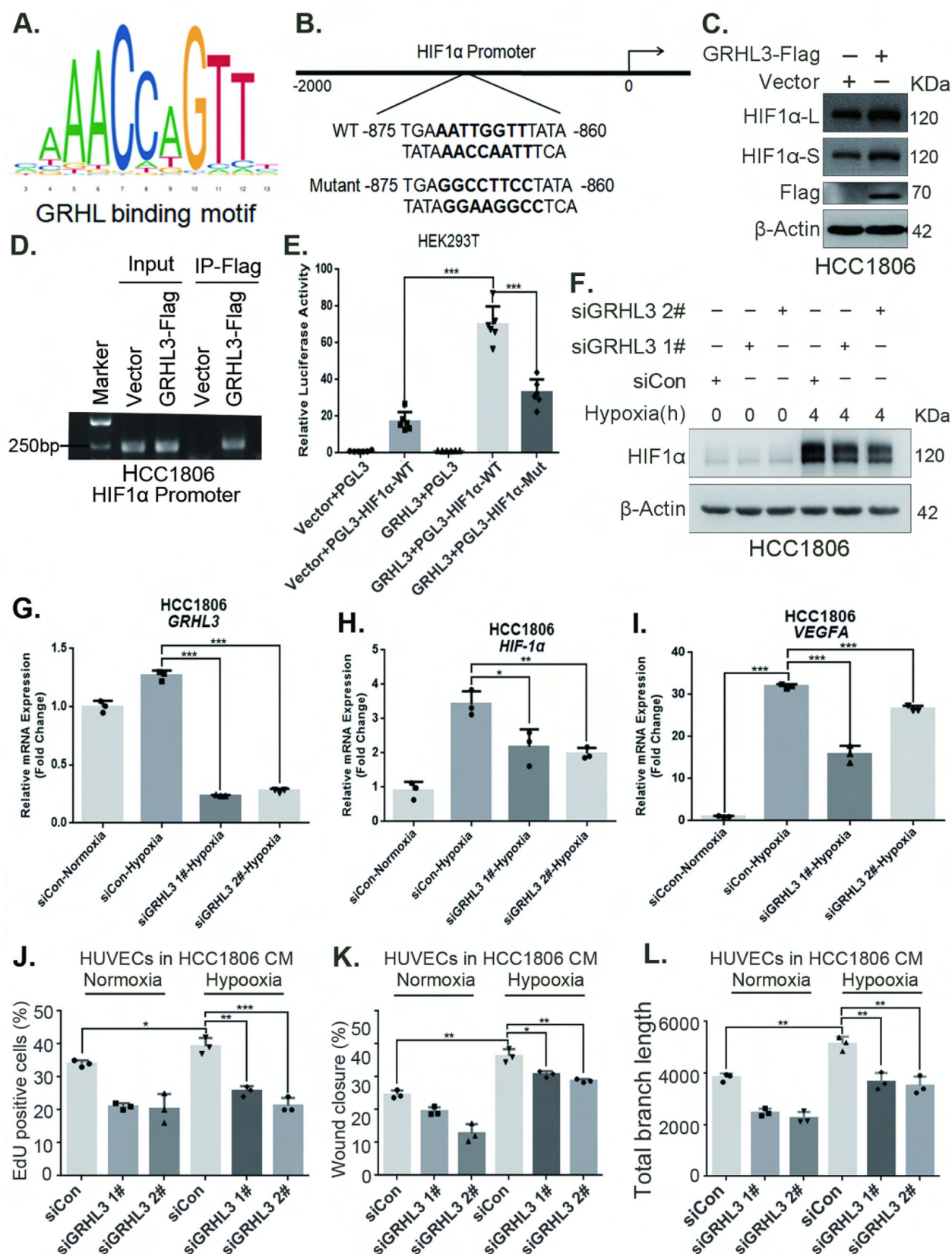


Figure 5.

GRHL3 enhances *HIF1A* transcription by binding to its promoter.

(A) The JASPAR website was used to predict the potential binding sequence of the transcription factor GRHL3 to the HIF1 α promoter. (B) Mutation pattern of the HIF1 α promoter binding sequence. (C) Overexpression of GRHL3 in HCC1806 cells stably overexpressing GRHL3 resulted in increased protein expression of HIF1 α . (D) ChIP-PCR experiments conducted in HCC1806 cells stably overexpressing GRHL3 revealed that GRHL3 could bind to the promoter region of the target gene *HIF1A*. (E) A luciferase assay performed in HEK293T cells indicated that GRHL3 could bind to the promoter sequence of *HIF1A* and increase its transcriptional activity. (F) In HCC1806 cells, GRHL3 was knocked down by siRNA, and the cells were subjected to hypoxia for 4 hours. Western blotting experiments revealed that the knockdown of GRHL3 downregulated the protein level of HIF1 α . (G-I) In HCC1806 cells, GRHL3 was knocked down by siRNA, followed by hypoxia treatment for 4 hours, after which RNA samples were collected. RT-qPCR experiments revealed that GRHL3 knockdown was effective and that the knockdown of GRHL3 could downregulate the mRNA levels of HIF1 α and its downstream target VEGFA. (J) In HCC1806 cells, GRHL3 was knocked down via siRNA, and the cells were then subjected to hypoxia for 24 hours. The conditioned medium was collected and used to treat HUVECs. EdU assays revealed that the conditioned medium after GRHL3 knockdown inhibited the proliferation of HUVECs. Statistical analysis of the EdU assay results was performed. (K) In HCC1806 cells, GRHL3 was knocked down via siRNA, and the cells were then subjected to hypoxia for 24 hours. The conditioned medium was collected and used to treat HUVECs. Wound healing assays revealed that the conditioned medium after GRHL3 knockdown inhibited the migration of HUVECs. Statistical analysis of the wound healing assay results was performed. (L) In HCC1806 cells, GRHL3 was knocked down via siRNA, and the cells were then subjected to hypoxia for 24 hours. The conditioned medium was collected and used to treat HUVECs. The results of the tube formation assay revealed that conditioned medium from GRHL3-knockdown HUVECs inhibited tube formation. Statistical analysis of the tube formation assay results was performed. *: $P < 0.05$, **: $P < 0.01$, ***: $P < 0.001$, t test.

Furthermore, conditioned medium from TNBC cells with GRHL3 knockdown inhibited HUVEC proliferation, migration, and tube formation (Fig. 5J-L and SFig. 4E-G). The overexpression of GRHL3 in TNBC cells activated the HIF1 α /VEGFA axis (Fig. 6A-B and SFig. 4H-J), resulting in increased proliferation, migration, and tube formation in HUVECs. This effect was reversed by knocking down HIF1 α in TNBC cells (Fig. 6C-E and SFig. 4K-M). Knockdown of GRHL3 (Fig. 6F) also inhibited tumor growth in HCC1806 xenograft mice (Fig. 6G-I) and reduced the number of microvessels in tumor tissues (Fig. 6J-K).

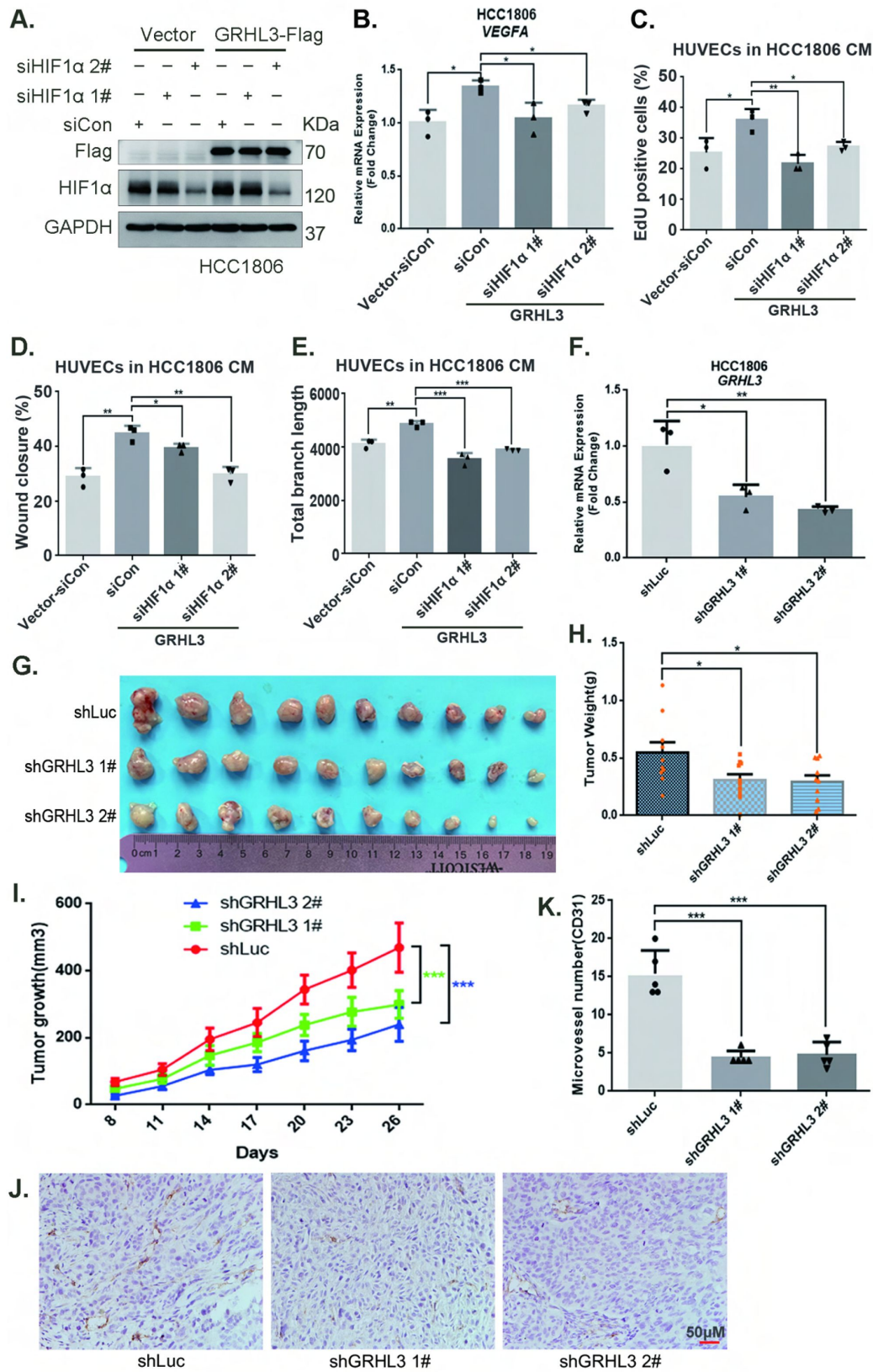


Figure 6.

GRHL3 enhances *HIF1A* transcription by binding to its promoter.

(A) In HCC1806 cells stably overexpressing GRHL3, HIF1 α was knocked down by siRNA, and the protein was subsequently collected. Western blotting experiments revealed that the overexpression of GRHL3 promoted the expression of HIF1 α , which could be rescued by the knockdown of HIF1 α . (B) In HCC1806 cells stably overexpressing GRHL3, HIF1 α was knocked down by siRNA, and then the RNA was collected. RT-qPCR experiments revealed that the overexpression of GRHL3 promoted the expression of the VEGFA mRNA, which could be rescued by the knockdown of HIF1 α . (C) HCC1806 cells stably overexpressing GRHL3 were used for the knockdown of HIF1 α via siRNA. The conditioned medium was then collected and used to treat HUVECs. EdU assays revealed that the overexpression of GRHL3 promoted the proliferation of HUVECs, which could be rescued by the knockdown of HIF1 α . (D) HCC1806 cells stably overexpressing GRHL3 were used for the knockdown of HIF1 α via siRNA. The conditioned medium was then collected and used to treat HUVECs. A wound healing assay revealed that the overexpression of GRHL3 promoted the migration of HUVECs, which could be rescued by the knockdown of HIF1 α . (E) HCC1806 cells stably overexpressing GRHL3 were used for the knockdown of HIF1 α via siRNA. The conditioned medium was then collected and used to treat HUVECs. The tube formation assay revealed that the overexpression of GRHL3 promoted the tube formation of HUVECs, which could be rescued by the knockdown of HIF1 α . (F) RT-qPCR was used to detect the knockdown of GRHL3 in HCC1806 cells. (G) Photographs of xenograft tumors from nude mice were taken to evaluate the role of GRHL3 in the growth of TNBC xenograft tumors. The results showed that the knockdown of GRHL3 significantly inhibited tumor growth. (H) The weights of the transplanted tumors from the nude mice were statistically analyzed, and the shGRHL3 1# and shGRHL3 2# groups were compared with the shLuc group. (I) The growth of transplanted tumors in nude mice was statistically analyzed, and the shGRHL3 1# and shGRHL3 2# groups were compared with the shLuc group. (J) An immunohistochemical assay was performed to detect the expression of the angiogenesis marker CD31 in xenograft tumors. (K) Statistical analysis of the number of microvessels in the immunohistochemical experiments. *: $P < 0.05$, **: $P < 0.01$, ***: $P < 0.001$, t test.

STAMBPL1 mediates *GRHL3* transcription by interacting with FOXO1

Our previous study demonstrated that STAMBPL1 is localized in the nucleus⁸. This protein may promote the transcription of GRHL3 through interactions with other transcription factors. Previous studies have indicated that FOXO1 acts as an upstream transcription factor of GRHL3¹⁴. Therefore, we aimed to investigate whether STAMBPL1 promotes GRHL3 transcription via FOXO1. Our experimental data revealed that knockdown of FOXO1 in TNBC cells not only inhibited the GRHL3/HIF1 α /VEGFA axis (Fig. 7A-D and SFig. 5A-D) but also reversed the stimulatory effects of STAMBPL1 on this axis (Fig. 7E-H and SFig. 5E-H). Furthermore, FOXO1 was found to bind to the promoter region of GRHL3 (Fig. 7I and SFig. 5I) and enhance its transcriptional activity (Fig. 7J). STAMBPL1 was shown to increase the transcriptional activation of FOXO1 at the GRHL3 promoter (Fig. 7K), and it also bound to the GRHL3 promoter; however, this interaction could be disrupted by FOXO1 knockdown (Fig. 7L).

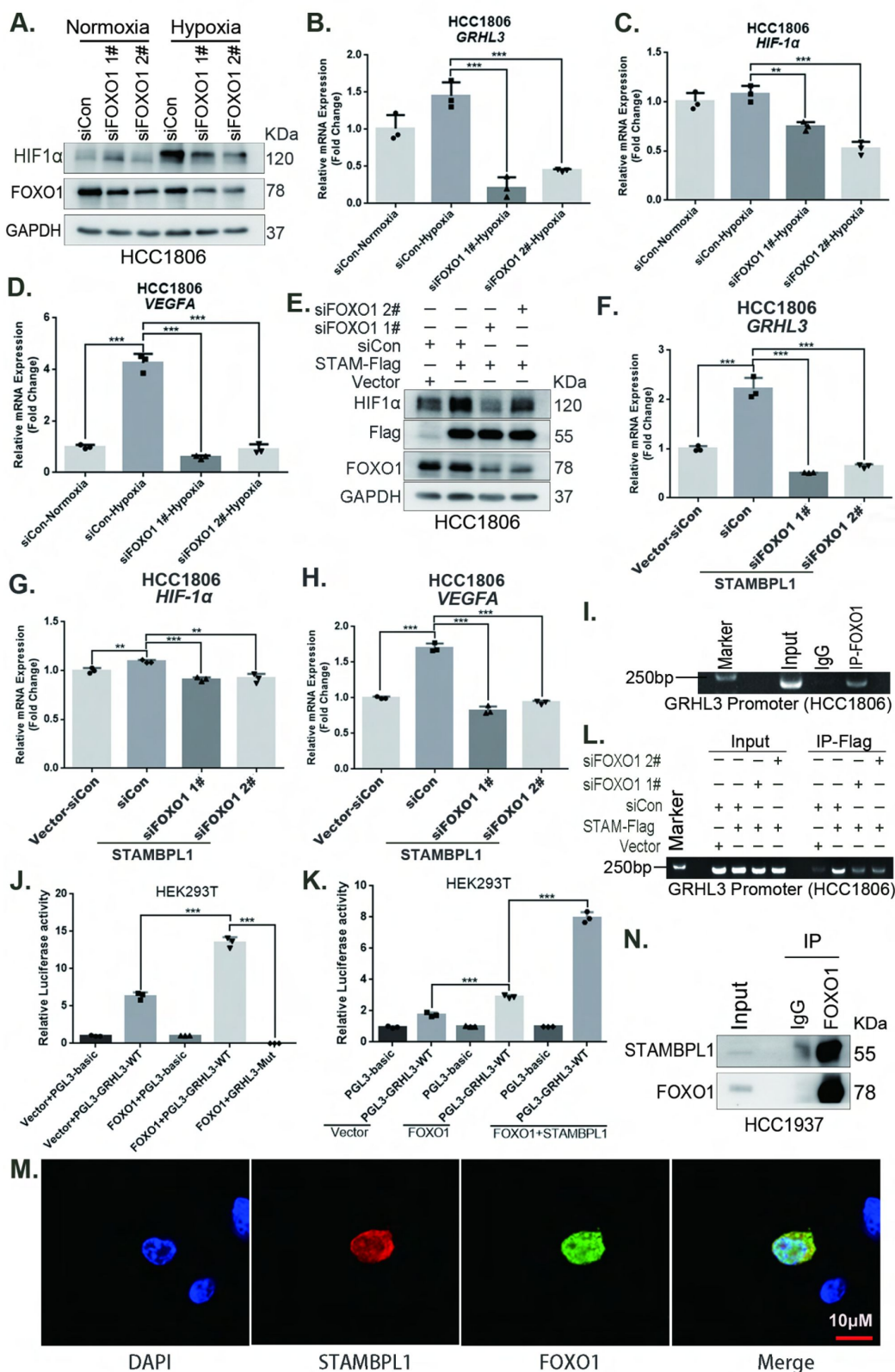


Figure 7.

STAMBPL1 mediates *GRHL3* transcription by interacting with FOXO1.

(A) Western blotting experiments revealed that the knockdown of FOXO1 in HCC1806 cells followed by hypoxia treatment for 4 hours inhibited HIF1 α protein expression. (B-D) RT-qPCR experiments revealed that the knockdown of FOXO1 in HCC1806 cells followed by hypoxia treatment for 4 hours decreased the mRNA levels of *GRHL3*/HIF1 α /VEGFA. (E) In HCC1806 cells stably overexpressing STAMBPL1, protein samples were collected after FOXO1 was knocked down via siRNA. Western blotting experiments revealed that the overexpression of STAMBPL1 promoted the expression of HIF1 α , which could be rescued by the knockdown of FOXO1. (F-H) In HCC1806 cells stably overexpressing STAMBPL1, RNA samples were collected after the knockdown of FOXO1 via siRNA. RT-qPCR experiments revealed that the overexpression of STAMBPL1 promoted the mRNA expression of *GRHL3*, HIF1 α and VEGFA, which could be rescued by the knockdown of FOXO1. (I) An endogenous ChIP-PCR assay was performed using an anti-FOXO1 antibody in HCC1806 cells. (J) A luciferase assay performed in HEK293T cells revealed that FOXO1 was able to bind to the promoter sequence of *GRHL3* and increase its transcriptional activity. (K) A luciferase assay conducted in HEK293T cells revealed that STAMBPL1 enhanced the activation of the *GRHL3* promoter by FOXO1. (L) ChIP-PCR experiments were performed in HCC1806 cells stably overexpressing STAMBPL1 after knocking down FOXO1 via siRNA. (M) PCDH-STAMBPL1-3 $\times$ Flag and PCDH-FOXO1-3 $\times$ Flag plasmids were cotransfected into HEK293T cells, and then, immunofluorescence experiments were performed. Red represents STAMBPL1 staining, green represents FOXO1 staining, and blue represents DAPI staining. (N) Endogenous STAMBPL1 protein was immunoprecipitated from HCC1806 cells via an anti-FOXO1 antibody. Immunoglobulin G (IgG) served as the negative control. Endogenous STAMBPL1 was detected via western blotting. *: $P < 0.05$, **: $P < 0.01$, ***: $P < 0.001$, t test.

To investigate the mechanism by which STAMBPL1 promotes *GRHL3* transcription through FOXO1, we conducted immunofluorescence and immunoprecipitation assays. We observed the colocalization of STAMBPL1 and FOXO1 in the nucleus (Fig. 7M) and identified an interaction between them in HCC1806 cells (Fig. 7N). Through the generation of a series of Flag-FOXO1/GST-fused STAMBPL1 deletion mutants, we mapped the regions of the proteins responsible for this interaction (SFig. 5F-G). The results of the GST pulldown assay indicated that the N-terminus (1–140 aa) of STAMBPL1 interacted with FOXO1 (SFig. 5H). Additionally, an immunoprecipitation assay revealed an interaction between the FOXO1 protein (250–596 aa) and STAMBPL1 (SFig. 5I). These findings suggest that STAMBPL1 promotes *GRHL3* transcription by interacting with FOXO1.

The combination of VEGFR and FOXO1 inhibitors synergistically suppresses TNBC xenograft growth

Through analysis of the TCGA database, we determined that *STAMBPL1*, *FOXO1*, and *GRHL3* exhibited high expression levels in TNBC tumors compared with non-TNBC tumors (Fig. 8A-C). To assess the potential therapeutic efficacy of cotreatment with the FOXO1 inhibitor AS1842856 and the VEGFR inhibitor apatinib *in vivo*, we conducted animal experiments in nude mice. HCC1806 cells overexpressing STAMBPL1 were orthotopically implanted into the mammary fat pads of 6-week-old female mice (n=16/group). Once the tumor volume reached approximately 50 mm³, the mice were divided into four subgroups to receive apatinib (50 mg/kg, once every two days), AS1842856 (10 mg/kg, once every two days), a combination of both drugs, or vehicle control for 20 days. Our findings indicated that STAMBPL1 overexpression enhanced breast cancer cell growth *in vivo*. While the individual inhibitory effects of the FOXO1 and VEGFR inhibitors on tumor growth were not significant, the combined treatment markedly suppressed tumor growth in nude mice (Fig. 8E-G). Importantly, the drug treatments did not affect the body weights of the mice (Fig. 8H). These results suggest that the combined administration of AS1842856 and apatinib effectively inhibits tumor growth in nude mice.

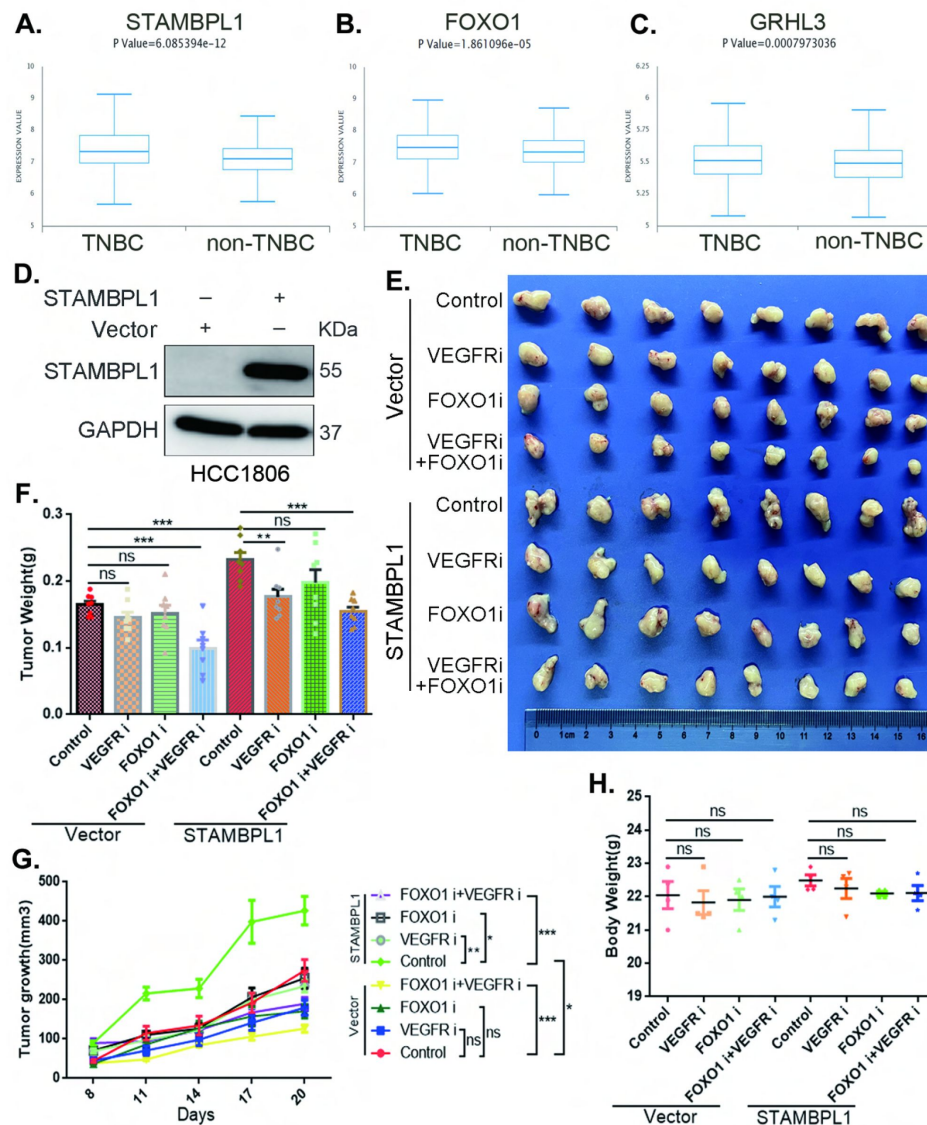


Figure 8.

The combination of VEGFR and FOXO1 inhibitors synergistically suppresses TNBC xenograft growth.

(A-C) TCGA database analysis revealed high expression levels of STAMBPL1, FOXO1, and GRHL3 in TNBC. (D) The effect of STAMBPL1 overexpression in HCC1806 cells was detected by Western blotting. (E) The effect of drug treatment on transplanted tumors in nude mice was evaluated by photographing the tumors. The overexpression of STAMBPL1 promoted tumor growth, whereas the combination of the FOXO1 inhibitor AS1842856 (10 mg/kg, every two days) and the VEGFR inhibitor apatinib (50 mg/kg, every two days) significantly inhibited tumor growth. (F-G) The weight and growth of the transplanted tumors in the nude mice were statistically analyzed. The vector control group and the STAMBPL1 overexpression group were compared. The nondrug group and the combined drug group in the vector control group were compared. The nondrug group and the combined drug group in the STAMBPL1 overexpression group were compared. (H) The final weights of the nude mice were statistically analyzed. *: $P < 0.05$, **: $P < 0.01$, ***: $P < 0.001$, t test.

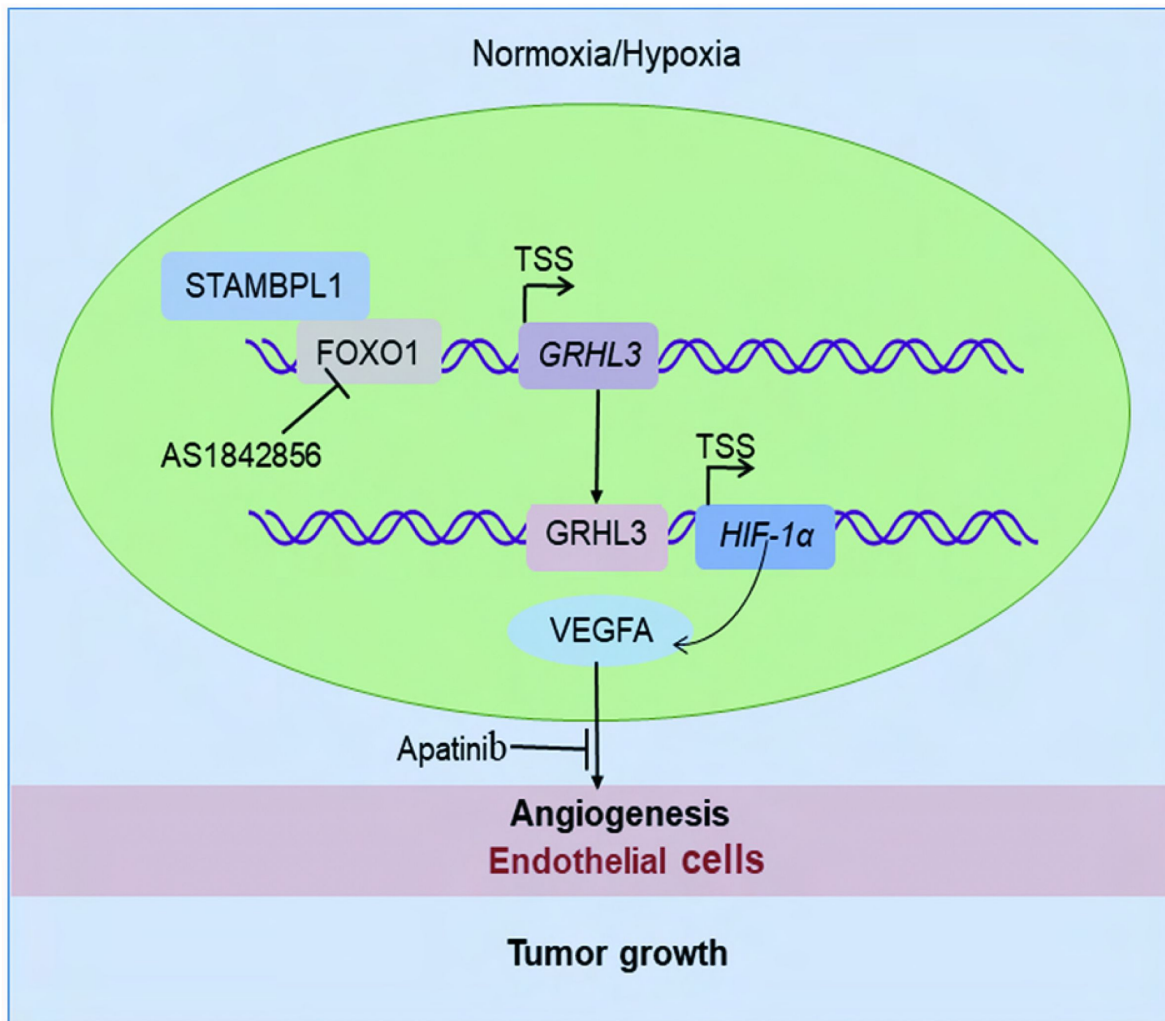


Figure 9.

The working model of this study.

STAMBPL1 interacts with FOXO1 to promote TNBC angiogenesis by activating the GRHL3/HIF1α/VEGFA axis.

Discussion

In this study, we discovered that STAMBPL1 promotes angiogenesis in TNBC by activating the HIF1 α /VEGFA pathway independently of its enzymatic activity. Through RNA-seq analysis, we revealed that STAMBPL1 positively regulates the transcription of *GRHL3*. Furthermore, we demonstrated that GRHL3 binds to the promoter of the *HIF1A* gene, thereby increasing its transcription. Additionally, we found that STAMBPL1 interacts with FOXO1 to facilitate the transcription of GRHL3/HIF1 α /VEGFA. Importantly, we confirmed that the combination of the FOXO1 inhibitor AS1842856 and the VEGFR inhibitor apatinib effectively inhibited tumor growth in nude mice. These findings suggest that both STAMBPL1 and FOXO1 may be potential therapeutic targets for inhibiting angiogenesis in TNBC. This study is the first to uncover the role of STAMBPL1 in promoting angiogenesis through the FOXO1/GRHL3/HIF1 α axis. Furthermore, a novel transcription factor, GRHL3, which regulates the transcription of HIF1 α , was identified. These findings provide valuable insights for developing new therapeutic strategies to target TNBC.

The role of STAMBPL1 in tumors has not been fully recognized. Recent studies have reported its involvement in the epithelial–mesenchymal transition of various cancers, and its absence has been shown to affect the mesenchymal phenotype of lung and breast cancer¹⁵. Our previous research demonstrated that STAMBPL1 can stabilize MKP1 through deubiquitination and that the deletion of STAMBPL1 and MKP1 increases the sensitivity of breast cancer cells to cisplatin¹⁶, suggesting that STAMBPL1 may be a potential therapeutic target for breast cancer. However, the mechanism by which STAMBPL1 activates the transcriptional activity of FOXO1 has not been elucidated, as STAMBPL1 does not stabilize the FOXO1 protein through its deubiquitinating enzyme activity. We speculate that STAMBPL1 may recruit other coactivators to FOXO1 transcription complexes. It will be important to develop a *Stambpl1* knockout mouse model to investigate the exact role of STAMBPL1 in TNBC. Additionally, we need to develop HIF1 α and FOXO1 antibodies suitable for immunohistochemistry to detect their expression in TNBC clinical samples.

Several studies have reported that FOXO1 inhibits tumor angiogenesis^{17–21}. Studies have shown that M2 macrophage-derived exosomal miR-942 promotes the migration and invasion of lung adenocarcinoma cells and facilitates angiogenesis by binding to FOXO1 to alleviate the inhibition of β -catenin, in which the upregulation of FOXO1 induces a decrease in cell invasion and angiogenesis *in vitro*¹⁷. FOXO1 inhibits gastric cancer growth and angiogenesis under hypoxic conditions via inactivation of the HIF1 α -VEGF pathway, possibly in association with SIRT1¹⁸. Cancer-associated fibroblast (CAF)-derived extracellular vesicles (EVs) deliver miR-135b-5p into colorectal adenocarcinoma cells to downregulate FOXO1 and promote HUVEC proliferation, migration, and angiogenesis¹⁹. Colorectal cancer cell-derived exosomes overexpressing miR-183-5p promote the proliferation, migration and tube formation of HMEC-1 (human microvascular endothelial cells) cells through the inhibition of FOXO1²⁰. Bladder cancer cell-derived exosomal miR-1247-3p facilitates angiogenesis by inhibiting FOXO1 expression²¹. However, the role of FOXO1 in breast cancer angiogenesis has not been studied, and our study revealed that FOXO1 can promote the expression of HIF1 α and VEGFA, suggesting that it may play a role in promoting angiogenesis in breast cancer.

Studies have demonstrated that the AKT–FOXO1 signaling pathway regulates the expression of GRHL3. To investigate this, the researchers utilized a previously published ChIP-seq dataset for FOXO1 from human endometrial stromal cells. They reported that FOXO1 occupied BRD4-bound enhancers near the *GRHL3* gene. The authors subsequently confirmed that the removal of EGF and insulin from the growth medium significantly increased the expression of GRHL3, but the

administration of the FOXO1 inhibitor AS1842856 partially blocked the induced expression of GRHL3¹⁴. These results suggest that FOXO1 plays a regulatory role upstream of GRHL3, and our study also confirmed that FOXO1 promotes the transcription of *GRHL3* by binding to its promoter.

GRHL3 is a highly conserved epidermal-specific developmental transcription factor that has recently gained attention in the field of cancer research²². To date, only a few genes regulated by GRHL3 have been identified. Studies have shown that GRHL3 levels are significantly reduced in human skin and head and neck squamous cell carcinomas and suggest that GRHL3 is a key tumor suppressor pathway in squamous cell carcinomas²³. GRHL3 was also found to be induced by TNF- α in the mammary carcinoma cell line MCF-7 and was identified as a TNF- α -induced endothelial cell migration factor with promigratory activity as high as that of VEGF²⁴. However, the specific role of GRHL3 in breast cancer is unclear. Studies have confirmed that GRHL3 strongly stimulates endothelial cell migration, which is consistent with an angiogenic, protumorigenic function²⁴. In our study, we demonstrated that GRHL3 promotes TNBC angiogenesis. HIF1 α , a known regulator of angiogenesis, is regulated by growth factors^{25,26}, cytokines²⁷ and mitogens^{28,29}. The EGFR/Akt pathway is a known positive regulator of HIF1 α , potentially through mTOR²⁵ or independent of it^{30,31}. While the mechanisms regulating HIF α protein expression have been extensively studied, those modulating HIF1 α transcriptional activity remain unclear. Our study reveals for the first time that GRHL3 promotes transcriptional activity by binding to the promoter of the *HIF1 α* gene.

Conclusions

In summary, STAMBPL1 promoted TNBC angiogenesis by activating the GRHL3/HIF1A pathway via FOXO1 in a non-enzymatic manner. These findings highlight the significant role of STAMBPL1 in TNBC angiogenesis and suggest that targeting the STAMBPL1/FOXO1/GRHL3/HIF1 α /VEGFA axis could be a potential therapeutic strategy to inhibit angiogenesis in TNBC.

List of abbreviations

- TNBC: triple-negative breast cancer
- ER: estrogen receptor
- PR: progesterone receptor
- HER2: human epidermal growth factor receptor 2
- EGFR: epidermal growth factor receptor
- PARP: poly (ADP-ribose) polymerase
- VEGFA: vascular endothelial growth factor A
- HUVECs: primary human umbilical vein endothelial cells
- CM: conditioned medium

Supplementary information

Supplementary Data (Supplementary Material 1) are available at JECCR Online.

Acknowledgements

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Authors' contributions

Ceshi Chen, Huifeng Zhang, Wenmin Cao and Huichun Liang: conceptualization; Huan Fang, Huichun Liang and Ceshi Chen: data curation; Huan Fang and Huichun Liang: formal analysis, methodology and writing-original draft with help from Chuanyu Yang, Dewei Jiang and Qianmei Luo; Ceshi Chen, Huichun Liang, Huifeng Zhang, and Wenmin Cao: funding acquisition, project administration, validation, supervision and writing-review & editing.

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Declarations

Ethics approval and consent to participate

All experimental procedures involving animals were performed in accordance to the Animal Welfare Law of China and the Guidelines for Animal Experiment of Kunming Institute of Zoology, Chinese Academy of Sciences.

Consent for publication

Not applicable.

Availability of data and materials

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Competing interests

The authors declare that they have no conflicts of interest.

Additional material

Supplementary Material 1

Figure S1. STAMBPL1 in TNBC cells enhances the activity of HUVECs. (A) HCC1937 cells were transfected with STAMBPL1 siRNA and subjected to hypoxia for 24 hours. The conditioned medium collected from these cells was then used to treat HUVECs. EdU assay revealed that the conditioned medium after STAMBPL1 knockdown inhibited the proliferation of HUVECs. (B)

Statistical analysis of the EdU assay results. **(C)** HCC1937 cells were transfected with STAMBPL1 siRNA and subjected to hypoxia for 24 hours. The conditioned medium collected from these cells was used to treat HUVECs. Wound healing assay demonstrated that the conditioned medium after STAMBPL1 knockdown inhibited the migration of HUVECs. **(D)** Statistical analysis of the wound healing assay results. **(E)** HCC1937 cells were transfected with STAMBPL1 siRNA and subjected to hypoxia for 24 hours. The conditioned medium collected from these cells was used to treat HUVECs. Tube formation assay showed that the conditioned medium after STAMBPL1 knockdown inhibited tube formation of HUVECs. **(F)** Statistical analysis of the tube formation assay results. *: $P < 0.05$, **: $P < 0.01$, ***: $P < 0.001$, t test.

Figure S2. STAMBPL1 in TNBC cells enhances the activity of HUVECs. (A and D) HCC1806 and HCC1937 cells were stably overexpressing STAMBPL1, and HIF1 α was knocked down using siRNA. The conditioned medium was then collected and used to treat HUVECs. EdU assay results demonstrated that the overexpression of STAMBPL1 promoted the proliferation of HUVECs, which could be rescued by knockdown of HIF1 α . Statistical analysis of the EdU assay results was performed. **(B and E)** HCC1806 and HCC1937 cells stably overexpressing STAMBPL1 were used to knock down HIF1 α using siRNA. The conditioned medium was collected and used to treat HUVECs. Wound healing assay results showed that the overexpression of STAMBPL1 promoted the migration of HUVECs, which could be rescued by knockdown of HIF1 α . Statistical analysis of the wound healing assay results was performed. **(C and F)** HCC1806 and HCC1937 cells stably overexpressing STAMBPL1 were used to knock down HIF1 α using siRNA. The conditioned medium was collected and used to treat HUVECs. Tube formation assay results showed that the overexpression of STAMBPL1 promoted the tube formation of HUVECs, which could be rescued by knockdown of HIF1 α . Statistical analysis of the tube formation assay results was performed. *: $P < 0.05$, **: $P < 0.01$, ***: $P < 0.001$, t test.

Figure S3. STAMBPL1 promotes *HIF1A* transcription via upregulating GRHL3. (A) RNA-seq result: the list of 18 genes downregulated after STAMBPL1 knockdown in HCC1806 cells. **(B)** RT-qPCR experiments showed that knockdown of STAMBPL1 inhibited GRHL3 mRNA levels in HCC1937 cells. **(C)** In HCC1937 cells stably overexpressing STAMBPL1, GRHL3 was knocked down using siRNA, and then the protein was collected. Western blotting experiments showed that overexpression of STAMBPL1 promoted the expression of HIF1 α , which could be rescued by knockdown of GRHL3. **(D-F)** In HCC1937 cells stably overexpressing STAMBPL1, GRHL3 was knocked down using siRNA and then the RNA was collected. RT-qPCR experiments showed that overexpression of STAMBPL1 promoted the mRNA expression of GRHL3, HIF1 α and its downstream target VEGFA, which could be rescued by knockdown of GRHL3. **(G)** Statistical analysis of EdU assay. **(H)** Statistical analysis of wound healing assay. **(I)** Statistical analysis of tube formation assay. *: $P < 0.05$, **: $P < 0.01$, ***: $P < 0.001$, t test.

Figure S4. GRHL3 enhances *HIF1A* transcription by binding to its promoter. (A) In HCC1937 cells, knockdown of GRHL3 using siRNA and treated with hypoxia for 4 hours. Western blotting experiments showed that knockdown of GRHL3 could down-regulate the protein level of HIF1 α . **(B-D)** In HCC1937 cells, GRHL3 was knocked down by siRNA, followed by hypoxia treatment for 4 hours, and then RNA samples were collected. RT-qPCR experiments showed that GRHL3 knockdown was effective, and knockdown of GRHL3 could down-regulate the mRNA levels of HIF1 α and its downstream target VEGFA. **(E)** In HCC1937 cells, GRHL3 was knocked down by siRNA and then treated with hypoxia for 24 hours, and then the conditioned medium was collected to treat HUVECs, EdU assay showed that the conditioned medium after knocking down GRHL3 inhibited the proliferation of HUVECs. **(F)** In HCC1937 cells, GRHL3 was knocked down by siRNA and then treated with hypoxia for 24 hours, and then the conditioned medium was collected to treat HUVECs. wound healing assay showed that the conditioned medium after knocking down GRHL3 inhibited the migration of HUVECs. **(G)** In HCC1937 cells, GRHL3 was knocked down by siRNA and then treated with hypoxia for 24 hours, and then the conditioned medium was collected to treat HUVECs, the tube formation assay showed that the conditioned medium after knocking

down GRHL3 inhibited tube formation of HUVECs. **(H)** In HCC1937 cells stably overexpressing GRHL3, HIF1 α was knocked down using siRNA, and then the protein was collected. Western blotting experiments showed that overexpression of GRHL3 promoted the expression of HIF1 α , which could be rescued by knockdown of HIF1 α . **(I)** In HCC1937 cells stably overexpressing GRHL3, HIF1 α was knocked down using siRNA and then the RNA was collected. RT-qPCR experiments showed that overexpression of GRHL3 promoted the mRNA expression of VEGFA, which could be rescued by knockdown of HIF1 α . **(J)** Statistical analysis of EdU assay results. **(K)** Statistical analysis of wound healing assay results. **(L)** Statistical analysis of tube formation assay results. *: $P < 0.05$, **: $P < 0.01$, ***: $P < 0.001$, t test.

Figure S5. STAMBPL1 mediates GRHL3 transcription by interacting with FOXO1. **(A)** Western blotting experiments showed that knockdown of FOXO1 in HCC1937 cells followed by hypoxia treatment for 4 hours inhibited HIF1 α protein expression. **(B-D)** RT-qPCR experiments showed that knockdown of FOXO1 in HCC1937 cells followed by hypoxia treatment for 4 hours inhibited the mRNA levels of GRHL3/HIF1 α /VEGFA. **(E)** The JASPAR website predicts the possible binding sequence of transcription factor FOXO1 to the GRHL3 promoter and its mutation pattern map. **(F)** The truncated structure diagram of STAMBPL1. **(G)** The truncated structure diagram of FOXO1. **(H)** The full-length and truncated plasmids of pEBG-STAMBPL1-GST and pCDH-FOXO1-no tag were transfected into HEK293T cells, and the proteins were collected for GST-pull down assay at 48 hours after transfection. **(I)** The full-length and truncated plasmids of pCDH-FOXO1-3 $\times$ Flag and pEBG-STAMBPL1-GST were transfected into HEK293T cells, and the proteins were collected for IP-Flag assay at 48 hours after transfection. *: $P < 0.05$, **: $P < 0.01$, ***: $P < 0.001$, t test.

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Author information

Huan Fang[#]

Kunming Institute of Zoology, Chinese Academy of Sciences, Kunming, China, Kunming College of Life Sciences, University of Chinese Academy of Sciences, Kunming, China

[#]These authors contributed equally to this work.

Huichun Liang[#]

Kunming Institute of Zoology, Chinese Academy of Sciences, Kunming, China

[#]These authors contributed equally to this work.

Chuanyu Yang

Kunming Institute of Zoology, Chinese Academy of Sciences, Kunming, China

Dewei Jiang

Kunming Institute of Zoology, Chinese Academy of Sciences, Kunming, China

Qianmei Luo

Kunming Institute of Zoology, Chinese Academy of Sciences, Kunming, China

Wenming Cao

The Department of Breast Medical Oncology, Zhejiang Cancer Hospital, Hangzhou, China

For correspondence: caowm@zjcc.org.cn

Huifeng Zhang

Department of Pharmacy, the First People's Hospital of Yunnan Province/the Affiliated Hospital of Kunming University of Science and Technology, Kunming, China

For correspondence: zhanghuifeng92@163.com

Ceshi Chen

Academy of Biomedical Engineering, Kunming Medical University, Kunming, China, The Third Affiliated Hospital, Kunming Medical University, Kunming, China, Kunming Institute of Zoology, Chinese Academy of Sciences, Kunming, China

ORCID iD: [0000-0001-6398-3516](https://orcid.org/0000-0001-6398-3516)

For correspondence: chenc@kmmu.edu.cn

Editors

Reviewing Editor

Yu Zhao

Institute of Radiation Medicine, Chinese Academy of Medical Sciences and Peking Union Medical College, Tianjin, China

Senior Editor

Caigang Liu

Shengjing Hospital of China Medical University, Shenyang, China

Reviewer #1 (Public review):**Summary:**

In this study by Fang et al., the authors show how STAMBPL1 promotes TNBC angiogenesis via a feed-forward GRHL3/HIF1a/VEGFA axis. They demonstrate that STAMBPL1 interacts with FOXO1, define the required domains in each protein, and illustrate that this interaction facilitates FOXO1 transcriptional factor activity, which then activating GRHL3/HIF1a/VEGFA signaling. Lastly, they show that the combination of VEGFR and FOXO1 inhibitors can synergistically suppress STAMBPL1-overexpressing TNBC.

Strengths:

The manuscript is clearly written, and the results are well explained. The observation that STAMBPL1 mediates GRHL3 transcription through its interaction with FOXO1 is novel. The findings also have important translational potential.

Weaknesses:

The mechanism by which STAMBPL1 mediates GRHL3 transcription through its interaction with FOXO1 is not sufficiently discussed, especially in relation to how STAMBPL1 regulates FOXO1. Some reported effects are modest.

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Reviewer #2 (Public review):**Summary:**

In their manuscript, Fang and colleagues make a notable contribution to the field of oncology, particularly in advancing our understanding of triple-negative breast cancer (TNBC). The research delineates the role of STAMBPL1 in promoting angiogenesis in TNBC through its interaction with FOXO1 and the subsequent activation of the GRHL3/HIF1A/VEGFA axis. The evidence presented is robust, with a combination of in vitro experiments, RNA sequencing, and in vivo studies providing a comprehensive view of the molecular mechanisms at play. The strength of the evidence is anchored in the systematic approach and the utilization of multiple methodologies to substantiate the findings.

Strengths:

The manuscript presents a methodologically robust framework, incorporating RNA-sequencing, chromatin immunoprecipitation (ChIP) assays, and a suite of in vitro and in vivo model systems, which collectively substantiate the claims regarding the pro-angiogenic role of STAMBPL1 in TNBC. The employment of multiple cellular models, conditioned media to assess HUVEC functional responses, and xenograft tumor models in murine hosts offers a comprehensive evaluation of STAMBPL1's impact on angiogenic processes. A salient strength of this work is the identification of GRHL3 as a transcriptional target of STAMBPL1 and the demonstration of a physical interaction between STAMBPL1 and FOXO1, which modulates GRHL3-driven HIF1A transcription. The study further suggests a potential therapeutic strategy by revealing the synergistic inhibitory effects of combined VEGFR and FOXO1 inhibitor treatment on TNBC tumor growth.

Weaknesses:

A potential limitation of the study is the reliance on specific cellular and animal models, which may constrain the extrapolation of these findings to the broader spectrum of human TNBC biology. Furthermore, while the study provides evidence for a novel regulatory axis involving STAMBPL1, FOXO1, and GRHL3, the multifaceted nature of angiogenesis may implicate additional regulatory factors not exhaustively addressed in this research.

Appraisal of Achievement and Conclusion Support:

The authors have successfully demonstrated that STAMBPL1 promotes HIF1A transcription and activates the HIF1 α /VEGFA axis in a non-enzymatic manner, leading to increased angiogenesis in TNBC. The results are generally supportive of their conclusions, with clear evidence that STAMBPL1 upregulates HIF1 α expression and enhances the activity of HUVECs. The study also shows that STAMBPL1 interacts with FOXO1 to promote GRHL3 transcription, which in turn activates HIF1A.

Impact on the Field and Utility:

This research is poised to exert a substantial impact on the oncological research community by uncovering the role of STAMBPL1 in TNBC angiogenesis and by identifying the STAMBPL1/FOXO1/GRHL3/HIF1 α /VEGFA axis as a potential therapeutic target. The findings could pave the way for the development of novel therapeutic strategies for TNBC, a subtype characterized by a paucity of effective treatment options. The methodologies utilized in this

study are likely to be valuable to the research community, offering a paradigm for investigating the role of deubiquitinating enzymes in oncogenic processes.

Additional Context:

It would be beneficial for readers to understand the broader context of TNBC research and the current challenges in treating this aggressive cancer subtype. The significance of this work is heightened by the lack of effective treatments for TNBC, making the identification of new therapeutic targets particularly important. Furthermore, understanding the specific mechanisms by which STAMBPL1 regulates HIF1 α expression could provide insights into hypoxia signaling in other cancer types as well.

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Reviewer #3 (Public review):

In this manuscript, Fang et al. describe a new oncogenic function of the STAMBPL1 protein in triple-negative breast cancer (TNBC). STAMBPL1 is a deubiquitinase that has been poorly studied in cancer. Previous reports identify it as a promoter of epithelial to mesenchymal transition or an inhibitor of cisplatin-induced cell death, but its participation to other cancer phenotypes has not been investigated. Fang et al. find that in cell line models of TNBC, STAMBPL1 promotes expression of the transcription factor HIF-1 α and its downstream target VEGF, with the consequence of stimulating neo-angiogenesis in vitro and in vivo.

Mechanistically, the authors find that this occurs via a non-enzymatic and indirect mechanism, that is by promoting the expression of GRHL3, a transcription factor that in turn binds to the HIF-1 α promoter to stimulate its transcription. Interestingly, the way by which STAMBPL1 promotes GRHL3 expression is by facilitating the transcriptional activity of FOXO1, a known regulator of GRHL3. Because the authors find that STAMBPL1 and FOXO1 interact, they suggest that STAMBPL1 may promote the formation of an active transcriptional complex containing FOXO1, perhaps by facilitating the recruitment of transcriptional coactivators. In conclusion, these data position for the first time the STAMBPL1 deubiquitinase in a FOXO-GRHL3 regulatory axis for the control of VEGF expression and tumor angiogenesis.

The main weaknesses of this work are that the relevance of this molecular axis to the pathogenesis of TNBC is not clear, and it is not clearly established whether this is a regulatory pathway that occurs in hypoxic conditions or independently of oxygen levels.

With respect to the first point, both FOXO1 and GRHL3 have been previously described as tumor suppressors, with reports of FOXO1 inhibiting tumor angiogenesis. Therefore, this work describes an apparently contradictory function of these proteins in TNBC. While it is not surprising that the same genes perform divergent functions in different tumor contexts, a stronger evidence in support of the oncogenic function of these two genes should be provided to make the data more convincing. As an example, the data in support of high STAMBPL1, FOXO and GRHL3 gene expression in TNBC TCGA specimens provided in Figure 8 is not very strong and it is not clear what the non-TNBC specimens are (whether other breast cancers or other tumors, perhaps those tumors where these genes perform tumor suppressive functions). To strengthen the notion that STAMBPL1, FOXO and GRHL3 are overexpressed in TNBC, the authors could provide a comparison with normal tissue, as well as the analysis of other publicly available datasets (like the NCI Clinical Proteomic Tumor Analysis Consortium as an example). Finally, it is not clear what are the basal protein expression levels of STAMBPL1 in the cell lines used in this study, as based on the data presented in Figures 2D and F it appears that the protein is not expressed if not exogenously overexpressed. It would be helpful if the authors addressed this issue and provided further evidence of STAMBPL1 expression in TNBC cell lines.

Linked to these considerations is the second major criticism, namely that it is not made clear if this new regulatory axis is proposed to act in normoxic or hypoxic conditions. The experiments presented in this paper are performed in both conditions but a clear

explanation as to why cells are exposed to hypoxia is not given and would be necessary being that HIF-1a transcription and not protein stability is being analyzed. Also, different hypoxic conditions are sometimes used, resulting in different mRNA levels of HIF-1a and its downstream targets and quite significant fluctuations within the same cell line from one experimental setting to the next. The authors should provide an explanation as to why experimental conditions are changed and, more importantly, the experiments presented in Figure 2 should be performed also in normoxia.

Another critical point is that necessary experimental controls are sometimes missing, and this is reducing the strength of some of the conclusions enunciated by the authors. As examples, experiments where overexpression of STAMBPL1 is coupled to silencing of FOXO1 to demonstrate dependency lack FOXO1 silencing the absence of STAMBPL1 overexpression. Because diminishing FOXO1 expression affects HIF-1a/VEGF transcription even in the absence of STAMBPL1 (shown in Figure 7C, D), it is not clear if the data presented in Figure 7G are significant. The difference between HIF-1a expression upon FOXO1 silencing should be compared in the presence or absence of STAMBPL1 overexpression to understand if FOXO1 impacts HIF-1a transcription dependently or independently of STAMBPL1.

In addition, some minor comments to improve the quality of this manuscript are provided.

(1) As a general statement, the manuscript is extremely synthetic. While this is not necessarily a negative feature, sometimes results are discussed in the figure legends and not in the main text (as an example, western blots showing HIF-1a expression) and this makes it hard to read thought the data in an easy and enjoyable manner.

(2) The effect of STAMBPL1 overexpression on HIF-1a transcription is minor (Figure 2) The authors should explain why they think this is the case and whether hypoxia may provide a molecular environment that is more permissive to this type of regulation.

(3) HIF-1a does not appear upregulated at the protein level protein by STAMBPL1 or GRLH3 overexpression, even though this is stated in the legends of Figures 2 and 6. The authors should show unsaturated western blots images and provide quantitative data of independent experiments to make this point.

In summary, adding necessary controls and performing additional experiments to substantiate the oncogenic function of these genes in TNCB would strengthen the authors' conclusions.

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